

## Report: Exercise 2

# Pattern Recognition on MNIST using SVM, MLP, and CNN

### Introduction :

In this exercise, we explore three widely-used approaches for pattern recognition:

- Support Vector Machines (SVM)
- Multilayer Perceptrons (MLP)
- Convolutional Neural Networks (CNN)

Using the full MNIST dataset, we aim to build, optimize, and compare these models to achieve the highest possible recognition accuracy.

To systematically evaluate each approach, we first implement and optimize a Support Vector Machine (SVM), a Multilayer Perceptron (MLP), and finally a Convolutional Neural Network (CNN).

For each model, we provide a brief overview of the approach, report the hyperparameter tuning process, present observations from training and validation performance, and finally report the test accuracy of the best-performing configuration.

### a) Support Vector Machine (SVM)

For this task, we trained a SVM to classify handwritten digits in the full MNIST dataset (60,000 training images, 10,000 test images)

Since MNIST has 10 classes, a multi-class SVM strategy was used (One-vs-One).

To determine the best kernel, we experimented with two types :

- Linear kernel
- RBF

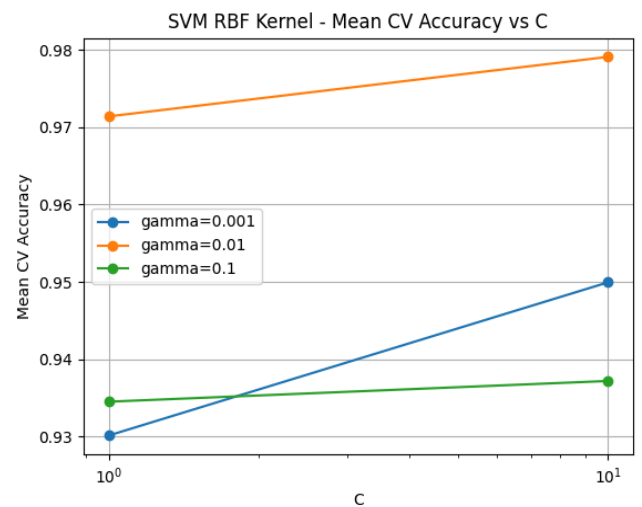
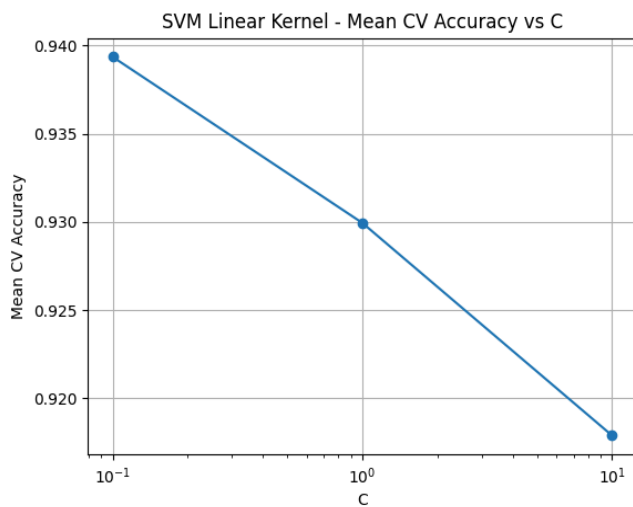
Hyperparameters were selected using 3-fold cross-validation on a split of the training set (50,000 for training, 10,000 for validation).

For hyperparameter tuning, the following grids were tested:

- Linear kernel:  $C=0.1, 1, 10$
- RBF kernel:  $C=1, 10$  and  $\gamma=0.001, 0.01, 0.1$

## Results :

### Cross-Validation Plots



Cross-validation results revealed different behaviors for the two kernels:

- Linear kernel: the accuracy has slightly decreased as C increased, indicating that a smaller C (more regularization) gave better generalization.
- RBF kernel: the accuracy has increased as C increased. The highest accuracy ( $\approx 0.98$ ) was achieved with  $C=10$  and  $\gamma=0.01$ , showing that both C and  $\gamma$  significantly affect performance.

The best hyperparameters were therefore:  $C=10$ ,  $\gamma=0.01$ , RBF kernel, resulting in a validation accuracy of 98.03% and a test accuracy of 98.22%.

| Kernel | C  | $\gamma$ | Validation Accuracy | Test Accuracy |
|--------|----|----------|---------------------|---------------|
| RBF    | 10 | 0.01     | 0.9803              | 0.9822        |

## Conclusion :

A well-tuned RBF SVM achieved excellent performance on MNIST, illustrating that non-linear kernels are effective for complex datasets, while linear kernels are suitable for simpler or roughly linearly separable cases.

### b) Multilayer Perceptron (MLP)

For this task, we trained a Multilayer Perceptron (MLP) to classify handwritten digits in the full MNIST dataset. Each image was flattened from  $28 \times 28$  pixels into a 784-dimensional vector.

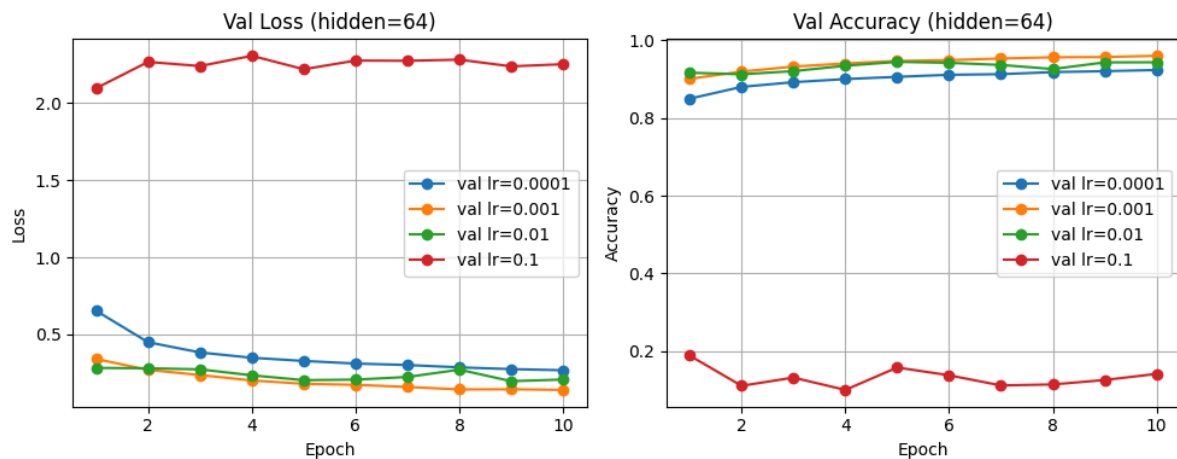
The MLP architecture consists of one hidden layer with a configurable number of neurons, followed by a ReLU activation, and a linear output layer with 10 neurons corresponding to the digit classes.

To optimize performance, we experimented with different hidden layer sizes [64, 128, 256] and learning rates [0.0001, 0.001, 0.01, 0.1]. The training set was split into 50,000 images for training and 10,000 images for validation. Each configuration was trained for 10 epochs.

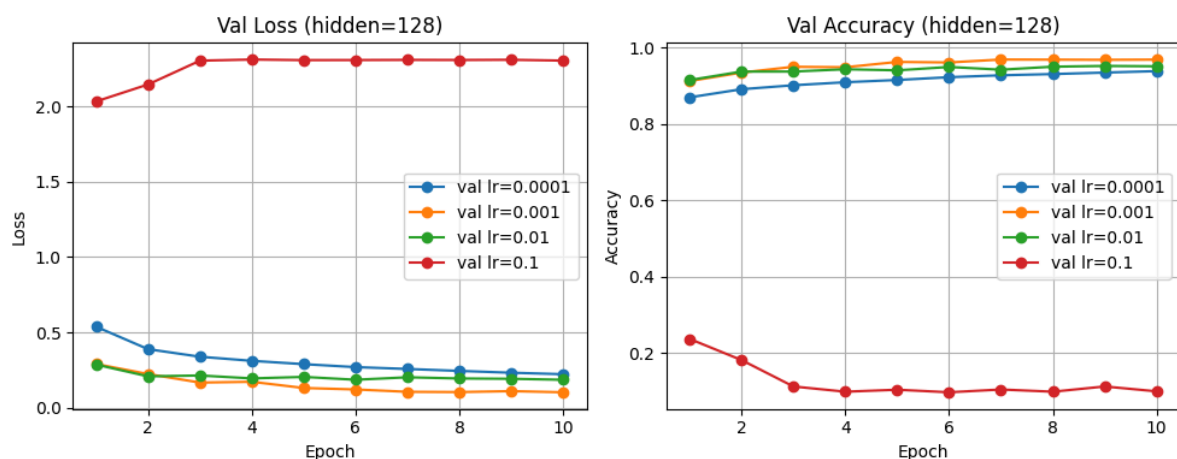
For each hyperparameter combination, we recorded training and validation loss and accuracy

## Results:

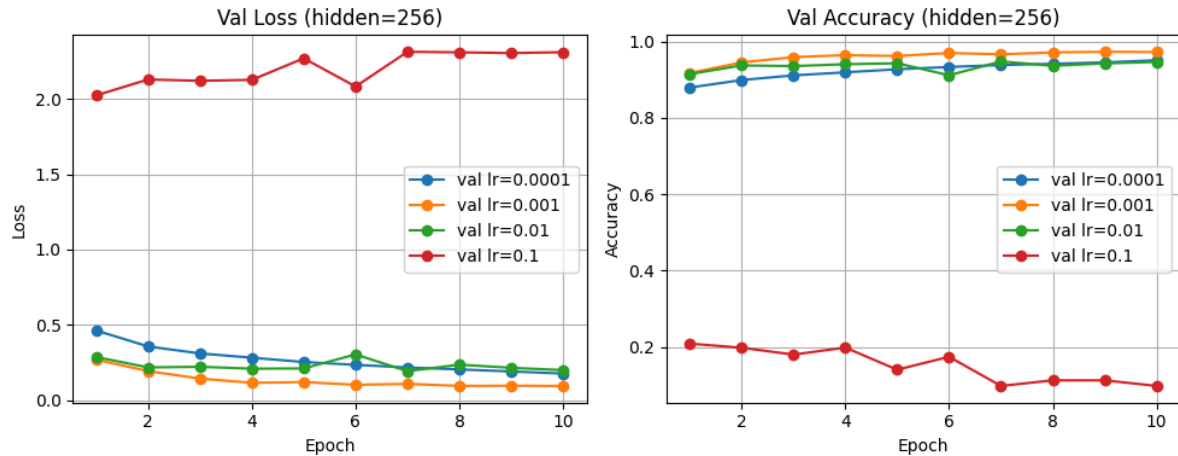
Validation plots :



- Validation Loss: Loss decreases for  $lr = 0.0001, 0.001, 0.01$  and stabilizes;  $lr = 0.1$  remains high.
- Validation Accuracy: Accuracy increases for smaller learning rates, highest for  $lr = 0.001$ ;  $lr = 0.1$  is significantly lower.



- Validation loss decreases for  $lr \leq 0.01$ ;  $lr = 0.1$  remains high.
- Validation accuracy is highest for  $lr = 0.001$ .



- Validation loss stabilizes at low values for  $lr \leq 0.01$ .
- Validation accuracy improves with hidden size 256, with  $lr = 0.001$  yielding the best performance

These results confirm that increasing the hidden layer size and using a moderate learning rate (0.001) provides the best balance between training stability and generalization, achieving strong performance on the MNIST test set.

So the best MLP configuration:

- Hidden size = 256
- Learning rate = 0.001
- Validation Accuracy = 0.9729
- Test Accuracy = 0.9743

### Conclusion:

The optimized MLP achieved strong performance on MNIST, reaching 97.43% test accuracy. Proper tuning of hidden size and learning rate is crucial for balancing learning stability and generalization

### c) Convolutional Neural Network (CNN)

For this task, we trained a Convolutional Neural Network (CNN) on the full MNIST dataset (60,000 training images and 10,000 test images).

The goal was to design an efficient architecture capable of accurately recognizing handwritten digits.

Since each image is  $28 \times 28$  pixels in grayscale, the model processes them through several convolutional and pooling layers before classification

To find the best configuration, we experimented with different network parameters:

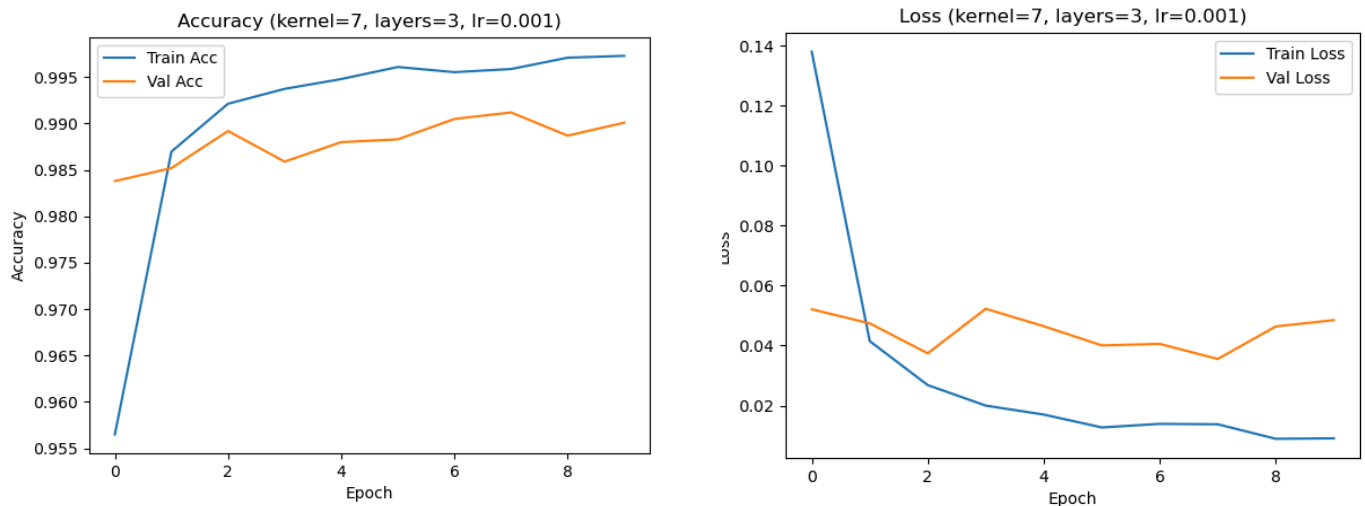
- Kernel sizes: 3, 5, 7
- Number of convolutional layers: 2, 3
- Learning rates: 0.0005, 0.001

- Optimizer: Adam
- Loss function: Cross-Entropy

The training set was split into 50,000 images for training and 10,000 images for validation. Each network was trained for 10 epochs, and the best model was selected based on the highest validation accuracy.

## Results:

Training and validation plots:



The plots below illustrate the evolution of training and validation performance over the epochs for different hyperparameter settings.

The training accuracy increased steadily, reaching around 0.995, while the validation accuracy stabilized slightly lower at approximately 0.990, showing that the model generalized well without significant overfitting.

Similarly, the training and validation losses both decreased and stabilized (around 0.01 and 0.04, respectively), confirming good convergence and learning stability.

So, after comparing all tested configurations, the best-performing CNN was obtained with:

- Kernel size = 7
- Number of convolutional layers = 3
- Learning rate = 0.001

This model achieved a validation accuracy of 99.01% and a test accuracy of 99.04%

| Kernel | Conv Layers | Learning Rate | Validation Accuracy | Test Accuracy |
|--------|-------------|---------------|---------------------|---------------|
| 7      | 3           | 0.001         | 0.9901              | 0.9904        |

**Conclusion:**

The optimized CNN achieved excellent accuracy ( $\approx 99\%$ ) on both validation and test sets, outperforming simpler models such as linear classifiers.

This demonstrates the strength of convolutional architectures in capturing spatial features and patterns from image data, making them highly effective for digit recognition tasks like MNIST

**Overall Conclusion:**

- The SVM achieved excellent results using a well-tuned RBF kernel, reaching 98.22% test accuracy, demonstrating the effectiveness of non-linear kernels for complex data.
- The MLP obtained 97.43% test accuracy when using an optimal hidden size of 256 and a moderate learning rate of 0.001, showing that careful tuning of architecture and learning rate is essential for stable training and good generalization.
- The CNN outperformed both SVM and MLP, achieving 99.04% test accuracy, highlighting the strength of convolutional architectures in capturing spatial features from images.

Overall, more complex architectures, such as CNNs, provide higher accuracy and better generalization for image recognition tasks. However, simpler models like SVMs and MLPs can still achieve competitive performance with proper hyperparameter tuning.